# Benchmark Radar: A Living Database and Search Engine for AI Benchmarks and Evaluation

Koutian Wu*[1,2] Junjie Zhou[3] Wenjun Zhang[4] Ergan Shang[5] Jiayu Wang[6] Pengqian Han[7] Junkai Wang[8] Wanghan Xu[9] Songyuanyi Lu[10] Rudy Mizrahi Celekli[11] Lazizbek Ravshanov[12] Yaoliang Bian[13,14] Lin Shi[15]

[1]Earth-Space-AI [2]Tacite AI [3]Hangzhou Dianzi University [4]The University of Queensland [5]Carnegie Mellon University [6]Xi'an Jiaotong University [7]The University of Auckland [8]Tsinghua University [9]Shanghai Jiao Tong University [10]The University of Hong Kong [11]Gradia [12]University of Cincinnati [13]Westlake University [14]Shenzhen Loop Area Institute [15]Cornell Tech

**Abstract**

Benchmark researchers and developers of large language models (LLMs) and other AI systems need to find relevant evaluations, locate their benchmark datasets and code, and understand the settings behind reported scores. We present Benchmark Radar, a living database and search engine for retrieval and discovery of AI benchmarks, covering LLM evaluation, agentic and tool-use benchmarks, coding, reasoning, safety, and domain-specific evaluations. The system combines daily discovery of benchmark papers, repositories, datasets, and releases with a searchable benchmark catalog, mentions in model cards and technical reports, and score histories. It retains source identities and citations so readers can inspect candidate benchmarks and their evaluation evidence. Daily discovery draws on 37 sources: 13 direct connectors and 24 first-party research and engineering feeds. The catalog contains 1,283 source records drawn from 4 benchmark catalogs and 12,916 numeric observations on 790 records. We describe collection and retrieval, audit the full catalog, and examine benchmark saturation, adoption trends, and the limits of score comparisons. A worked example walks through a complete prior-art search, showing how to query the catalog and inspect benchmark evidence when designing a new evaluation. We release the web dashboard with a benchmark leaderboard, a Pareto frontier view of score against measured use, saturation and trend views, daily feeds, downloadable evidence, a command-line interface (CLI) for offline queries, and reproducible analysis.

*Corresponding author: k@tacite.ai

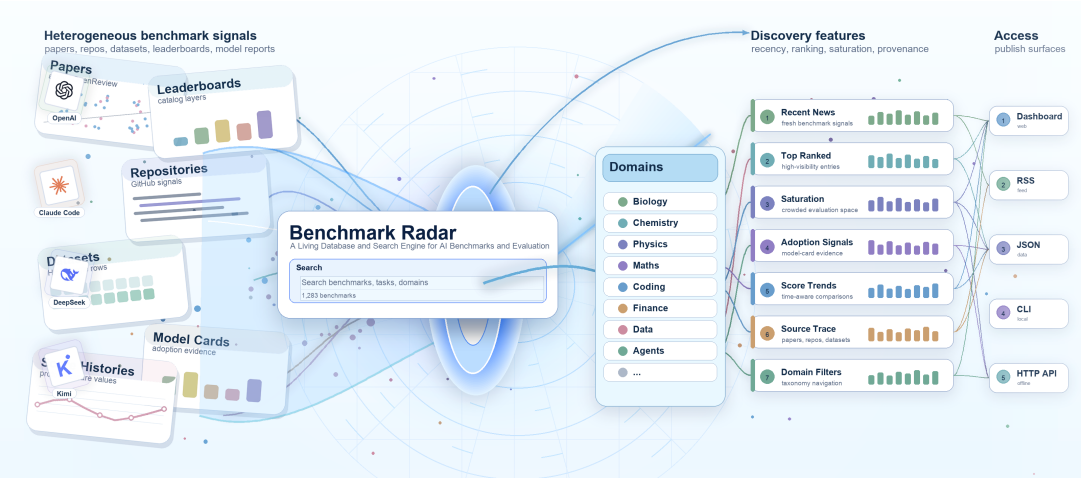



Figure 1: Benchmark Radar connects discovery, retrieval, and source inspection.

# 1 Introduction

The Transformer architecture established attention-based sequence modeling as a foundation for modern large language models (LLMs) [66]. GPT-6 Astra and Claude Fable 5.1 are designed for coding, research, and tasks that span multiple tools [3, 46]. Comparing these models requires benchmarks that expose failures and distinguish capability gains from changes in prompts, data splits, or evaluation procedures [21, 49, 52, 57].

Evaluation also matters beyond general-purpose chat. In recommender systems, Transformer-style sequence models and LLM-based methods support user, item, and preference modeling, with efficient training needed at production scale [29, 55, 63]. Foundation models are also used to model biological, geoscientific, and physical processes [13, 56, 65, 69, 77]. In social and political analysis, language models support text annotation and simulated human samples, while network methods quantify balance and polarization dynamics [4, 16, 35, 36, 45, 58]. Benchmark scores inform claims of progress across these scientific, industrial, and decision-support settings.

Benchmarks differ in the abilities they test. Broad evaluations of knowledge and reasoning include MMLU [21], GPQA [52], and Humanity's Last Exam [49]. Others isolate more specific capabilities: SWE-bench measures whether models can resolve real GitHub issues [27], LiveCodeBench tests code generation with date-based splits to reduce contamination from training data [26], and Terminal-Bench and Long-Horizon-Terminal-Bench test command-line agents on operational tasks [37, 41]. The Harbor framework ports more than 80 such benchmarks to a common execution contract so that arbitrary agents can be evaluated against them, and its Harbor-Index curates 82 difficult tasks drawn from 29 of those benchmarks [20, 59]. Domain-specific evaluations include FinanceBench for finance and economics [25], MATH for mathematics [22], SciBench for scientific problem solving [67], LAB-Bench for biology [34], and ChemBench for chemistry [42]. ESM-Bench tests agents' understanding of Earth system model physics and code [70]; ResearchClawBench evaluates end-to-end autonomous scientific research [74]; and ASI-Bench examines scientific exploration and execution with progressively less methodological guidance [78]. ScienceIDE converts scientific code repositories into executable, agent-learnable environments for training and evaluating scientific agents [15]. Benchmark construction is itself being automated: Benchmark Agent builds benchmarks end to end, from query analysis and subtask design to annotation and quality control [73]. Their tasks and scoring rules specify which model behaviors count as successful performance.

Finding a benchmark, its dataset or repository, and reports of its use requires searching paper servers, code hosting sites, dataset hubs, vendor releases, blogs, and benchmark catalogs. Existing resources such as LLM Stats, OpenCompass, and Artificial Analysis provide leaderboards, evaluation platforms, and model-analysis views [5, 39, 47]. Connecting newly released benchmarks to their task materials and later use in model reports requires consulting several separate resources. Benchmark Radar addresses this gap by combining catalog retrieval with daily discovery, mentions in model reports, and scores with documented evaluation settings and sources.

Benchmark Radar gathers benchmark-related artifacts from public sources, groups observations with matching identifiers, and exposes a searchable index with source labels, dates, benchmark mentions, and scores (Figure 1). We present the search system, describe the collection and ranking methods needed to reproduce its outputs, and examine benchmark coverage, documentation and scored-model coverage across all catalog sources, source concentration, and gaps in evaluation settings.[1]

The system preserves one record per source benchmark, including entries without scores, dates, or citations. Reviewed identity links connect related records while retaining their separate measurements. We contribute this living search system, shared web and offline access to its evidence, and a repro-

[1]Source code: https://github.com/ktwu01/benchmark-radar.

ducible full-catalog census. Section 5 shows how a contributor used retrieval and source inspection to assemble prior art.

## 2 Related Work and Scope

**Benchmark catalogs and evaluations.** LLM Stats publishes benchmark descriptions and reported model results [39]. OpenCompass provides an evaluation platform and a benchmark registry with artifact metadata [47]. Artificial Analysis publishes model evaluations and their methodology [5]. Benchmark Radar uses records from these sources alongside model-report evidence. Benchmark Radar adds daily discovery, artifact histories, benchmark retrieval, and source inspection through shared web and offline access, helping readers investigate evaluations across the contributing catalogs.

**Documenting evaluation evidence.** Model cards and datasheets motivate documenting evaluation conditions and dataset characteristics [14, 43]. Benchmark Radar retains that information with its citations. Benchmarks such as GPQA [52], SWE-bench [27], and SciBench [67] define different tasks. The census measures which evidence each record provides.

**Evaluating benchmarks themselves.** Recent work examines benchmark saturation, item quality, and the interpretation of aggregate scores. In a systematic study of 60 language model benchmarks, Akhtar et al. [1] define saturation and report that nearly half exhibit it, with prevalence increasing with benchmark age. They find that expert curation affects resilience to saturation, and that public test data has no measurable effect. Sample-level auditing of MMLU, ARC, WinoGrande, HellaSwag, and TruthfulQA identifies within-benchmark variation obscured by aggregate accuracy [60]. A reference-free judging framework assesses conversational-agent benchmarks on consistency, complexity, and policy coverage [33]. For safety benchmarks developed for larger models, rankings of smaller models vary with the treatment of ambiguous responses [54]. Inverse-density weighting reduces the influence of benchmark multiplicity on aggregate scores [38]. Gilda and Gilda [17] argue that evaluation scores should state their evidential scope and validity window, with conservative aggregation when component signals differ in reliability.

Benchmark Radar complements these methods by making benchmark records, available measurements, and provenance searchable. We used its command-line client to identify candidate papers for this section, supplemented the results with coauthor recommendations, and reviewed the source papers.

## 3 System and Methods

### 3.1 System Overview and Daily Discovery

Benchmark Radar adapts BuilderPulse's daily public-source collection approach [72]. Figure 2 separates the benchmark catalog from discovery history. Model reports include model cards, technical reports, system cards, and release posts; they contribute through the same record structure as benchmark registries.

Table 1 summarizes the four catalog sources and their primary uses in the v0.11.0 release.

| Catalog source | Rows | Primary use |
|---|---|---|
| LLM Stats | 687 | Benchmark names and descriptions with source provenance. |
| OpenCompass Hub | 461 | Paper, repository, release, and dataset links with source-specific identity fields. |
| Artificial Analysis | 25 | Current commercial evaluation catalog entries. |
| Model reports | 110 | Benchmarks mentioned in model reports, including records without scores. |
| Total | 1,283 | All four sources, retaining each record's source label. |

Table 1: Catalog sources in the v0.11.0 release. Rows count source-specific benchmark records, including records without scores.

### 3.2 Discovery Collection

Daily discovery observations describe mentions, releases, and updates found across public sources. An observation is one collected record; an artifact is a paper, repository, dataset, release, or page linked by exact identifiers. We retain their histories alongside the catalog and count them separately from benchmark records.

Each collection run searches a 48-hour window, records counts and errors by source, removes future-dated rows, and publishes the daily snapshot once all core sources are healthy. At the cutoff, arXiv [6], Hugging Face Hub [24], and GitHub Search [19] were the core sources. Across 13 direct connectors and 24 first-party feeds, additional routes cover scholarly indexes, dataset hosts, repository releases, and institutional feeds. Appendix D records their cutoff status.

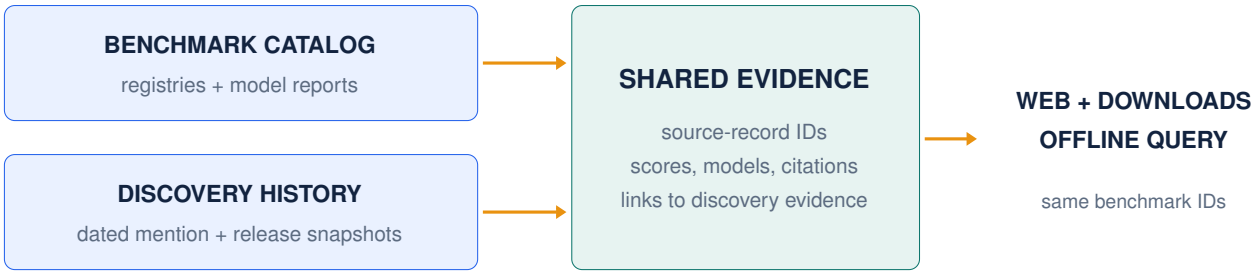



Figure 2: Two input paths feed the published outputs: dated discovery snapshots preserve mention and release evidence, while benchmark registries and model reports populate the shared catalog. Clients use the catalog's stable source-record IDs.

Table 2 describes the interfaces. Search and exports retain the full catalog; display filters apply to the current view.

| Interface | Reader question | Evidence and scope |
|---|---|---|
| Today | What appeared or changed? | Daily discovery observations, source health, and links. |
| Search | Which benchmarks match this task? | The complete catalog, including unscored records. |
| Leaderboard | Which evaluations combine lower scores with broader measured use? | Scored source records; distinct scored models or cited documents supply the selected count. |
| Saturation | What scores and settings were reported? | Benchmark browsing and complete score histories; search reaches every source and year. |
| Blog | Can I read and share a daily brief? | Daily briefs and an archive built from committed discovery snapshots. |
| CLI | Can I inspect the same evidence locally? | The offline query client uses the same benchmark IDs and stable JSON response format. |

Table 2: Reader questions and evidence scope.

### 3.3 Preserve Records Before Comparing Measurements

A source record is one benchmark entry from one contributing source. We normalize names, identifiers, artifact links, score observations, model identities, and cited documents into common fields. A record remains in the catalog if a score, date, or citation is absent. Reviewed identity links connect related records while preserving their separate observations and counts.

Daily discovery contributes a different kind of evidence: collected mentions, releases, and updates. Exact identifiers such as DOIs, arXiv IDs, and repository URLs link those observations to artifacts. Appendix C records detailed collection settings and display filters; Appendix D records source health at the cutoff.

### 3.4 Retrieve Candidates with Their Evidence from the Dashboard or the CLI

The same benchmark IDs reach web search, detail pages, dataset exports, and offline clients. Lexical search uses BM25F, a field-weighted word-matching score [53], with bounded boosts for name and phrase matches. Each result exposes matched and missing query words, the fields they occur in, and the score components. A shared query service supplies the CLI and HTTP interfaces with the same response format and local data provenance.

The CLI keeps its own copy of the data. `benchmark-radar init` downloads the dataset archives and `benchmark-radar sync` updates them; a local manifest verifies each archive's SHA-256 checksum and records its schema version and provenance. Queries then run without network access, so a search can be repeated against a recorded dataset version. A search covers the normalized source records (`catalog`), the daily discovery snapshots (`radar`), or both (`all`). Filters restrict results by modality, openness, source, and the presence of a paper, repository, or dataset link. `benchmark-radar serve` answers the same queries on a local HTTP endpoint for programmatic use.

Commands that return results accept `--json`, which prints a versioned payload holding the ranking explanation above, the matching policy applied, and the version and build date of the local data. The same interface ships as an agent skill (`npx skills add ktwu01/benchmark-radar`) that states when a request calls for a benchmark query, which queries to run, and how to install the CLI and initialize

its data when they are absent. A reader and an agent then work from the same records and the same score components.

An analyst or agent can try focused query variants, inspect a record's tasks and score settings, and follow its citations. The interface preserves the evidence needed to judge a candidate's suitability.

### 3.5 Audit the Full Population

The census starts with every record in the rebuilt catalog index and reads its detail file. We count finite numeric score observations once by observation ID. Within each benchmark record, we count scored models by source model ID, preserving separately evaluated configurations, and cited documents by document ID. The global model count uses the registry's recorded identity links.

Eligibility depends on the measurement a calculation needs. A declared percentage unit, known score direction, and numeric values within 0–100 permit a percentage-scale summary. Percentage units come from the source's own declaration; display multipliers are recorded separately.

Date coverage counts valid recorded benchmark release dates. Score entries retain their own date basis, including model announcements and document publication; both serve as proxies for the evaluation date. Appendix F records the software revision and links to the input hashes, census script, and validation procedure.

## 4 Results

Appendix A provides the complete linked census; Appendix B details document, model, score, and date coverage.

### 4.1 Catalog Coverage and Task Materials

The rebuilt catalog contains 1,283 source records across 4 sources. We found 12,916 numeric score observations on 790 records, with 493 records lacking numeric scores (Table 3).

| Source | Records | Scored | Unscored | Numeric scores |
|---|---|---|---|---|
| LLM Stats | 687 | 679 | 8 | 5,544 |
| OpenCompass Hub | 461 | 0 | 461 | 0 |
| Artificial Analysis | 25 | 25 | 0 | 7,050 |
| Model reports | 110 | 86 | 24 | 322 |
| **Total** | **1,283** | **790** | **493** | **12,916** |

Table 3: Score coverage across the full catalog: 790 records have numeric observations and 493 do not. Scores count numeric observations.

Of the 493 unscored records, 464 have at least one paper, repository, or dataset link. In the full catalog, 475 records link to papers, 506 to repositories, and 293 to datasets. For example, the OpenCompass Hub record for A-Bench links a paper, code repository, and dataset despite having no archived score.

For scored records, Figure 3 lets readers browse by date, reported score, and number of scored models. Figure 4 shows the reported model scores within one source record.

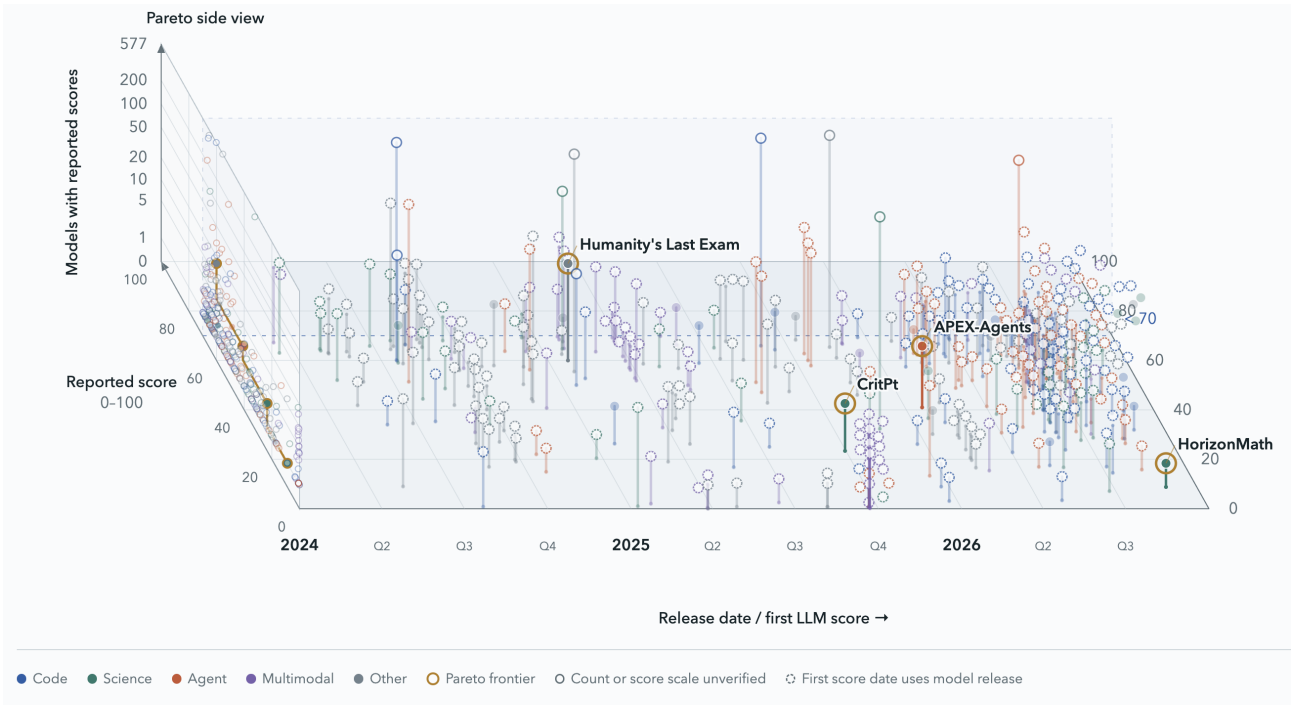



Figure 3: Benchmark Frontier on the Leaderboard page. The view combines reported scores, scored-model counts, and benchmark release dates or first-score date proxies. Gold rings mark the interface's Pareto candidates; hollow marks flag unverified scales or counts, and dotted outlines flag model-release date proxies. The image links to the interactive view.

## 4.2 Benchmark Taxonomy Across the Full Catalog

Most source records carry a capability label. We classify 1,279 of 1,283 records into 11 top-level domains and 63 sub-domains (Figure 5). Labels come from the publishers' own fields—OpenCompass Hub dimensions, LLM Stats categories, Artificial Analysis categories, and the model-report registry domains. The remaining 4 records are LLM Stats community rows whose crawl supplied no description, category, or modality; the catalog records that as the reason.

Interaction paradigm and input modality are recorded as *facets*: properties held beside a record's domain, so every record carries exactly one Level 1 class and any number of facet values. One record can count toward several facets.

Separating the axes matters: a benchmark that resolves repository issues is a coding benchmark run as an agent. Of the 345 agentic records, 128 take Agentic & Tool Use as their Level 1 class and 117 sit under Coding & Software Engineering, with the rest spread across 6 further classes. A scheme with one axis has to choose, and choosing the domain hides those 117 records from any count of agentic evaluation.

Figure 6 puts each of the 615 records with a benchmark release date on its release year. The 668 records without a release date stay in a separate column with the same vertical scale. Panel B reports shares for years with at least 30 dated records and a source mix near the pooled mix. Reweighting each of these years to a common source mix changes the agentic share by at most 1.1 percentage points.

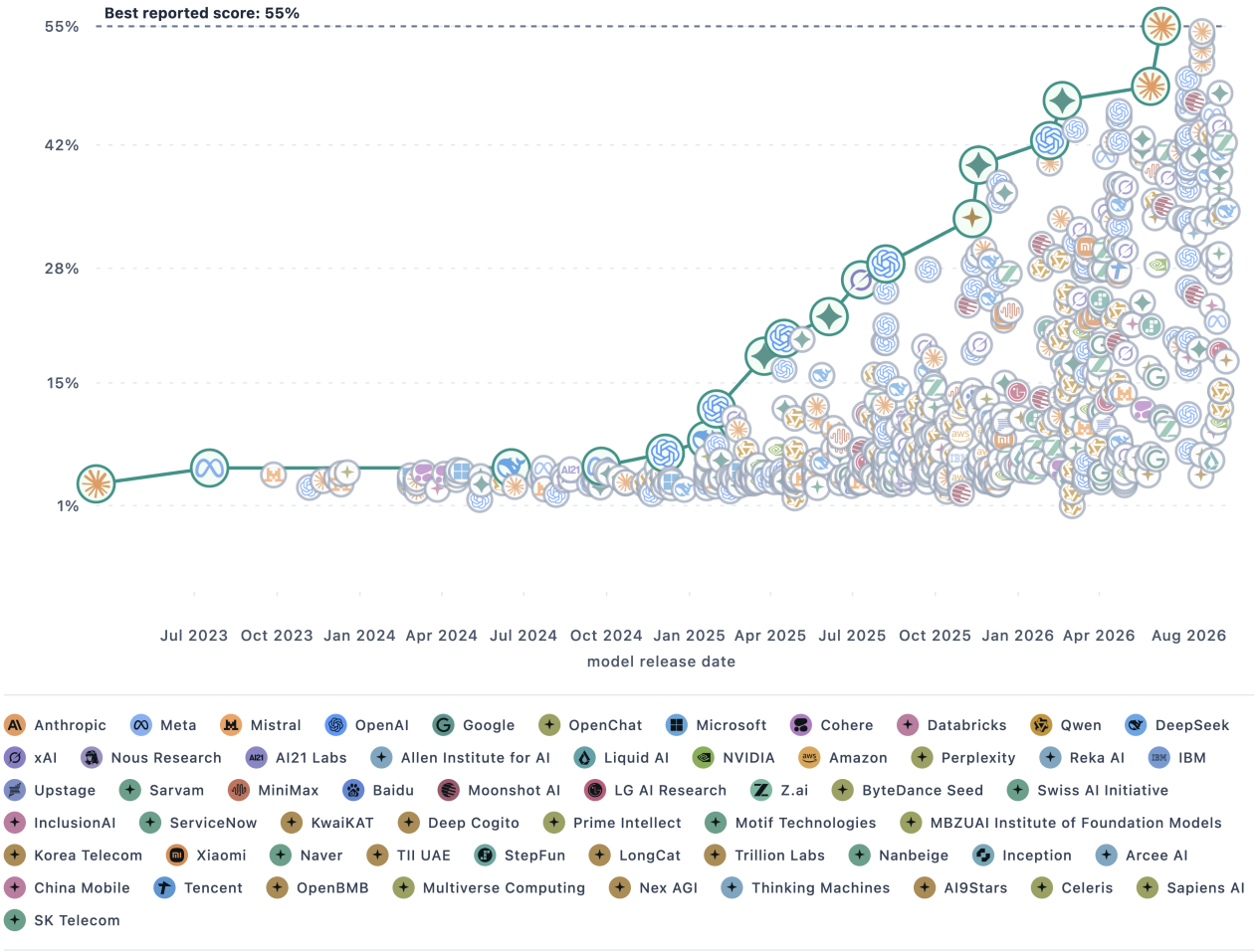



Figure 4: Inspecting Humanity's Last Exam in Benchmark Radar. This Artificial Analysis record contains 577 reported scores; the highest displayed score is 55.47. Marks identify model organizations, and the line connects successive best scores ordered by model release date. The image links to the interactive view.

### 4.3 Discovery Coverage Alongside the Catalog

The daily discovery collection contains 11,068 observations and 6,546 artifacts linked by exact identifiers across 46 snapshots. 4 snapshots are simulated historical backfills; their dates describe reconstructed collection windows.

Five discovery source labels account for 9,743 of 11,068 observations (88.0%). Figure 7 includes the remaining labels in one aggregate bar. Source caps and collection failures shape this mix. Of the 6,546 artifacts, 59 have observations from multiple sources. Cross-source matching relies on exact identifiers, so artifacts with missing or inconsistent identifiers stay unmatched.

Figure 8 complements source coverage with the daily reading view. Its category cards separate new releases from updates, while the chart exposes changes in surfaced evidence and attention alongside collection failures.

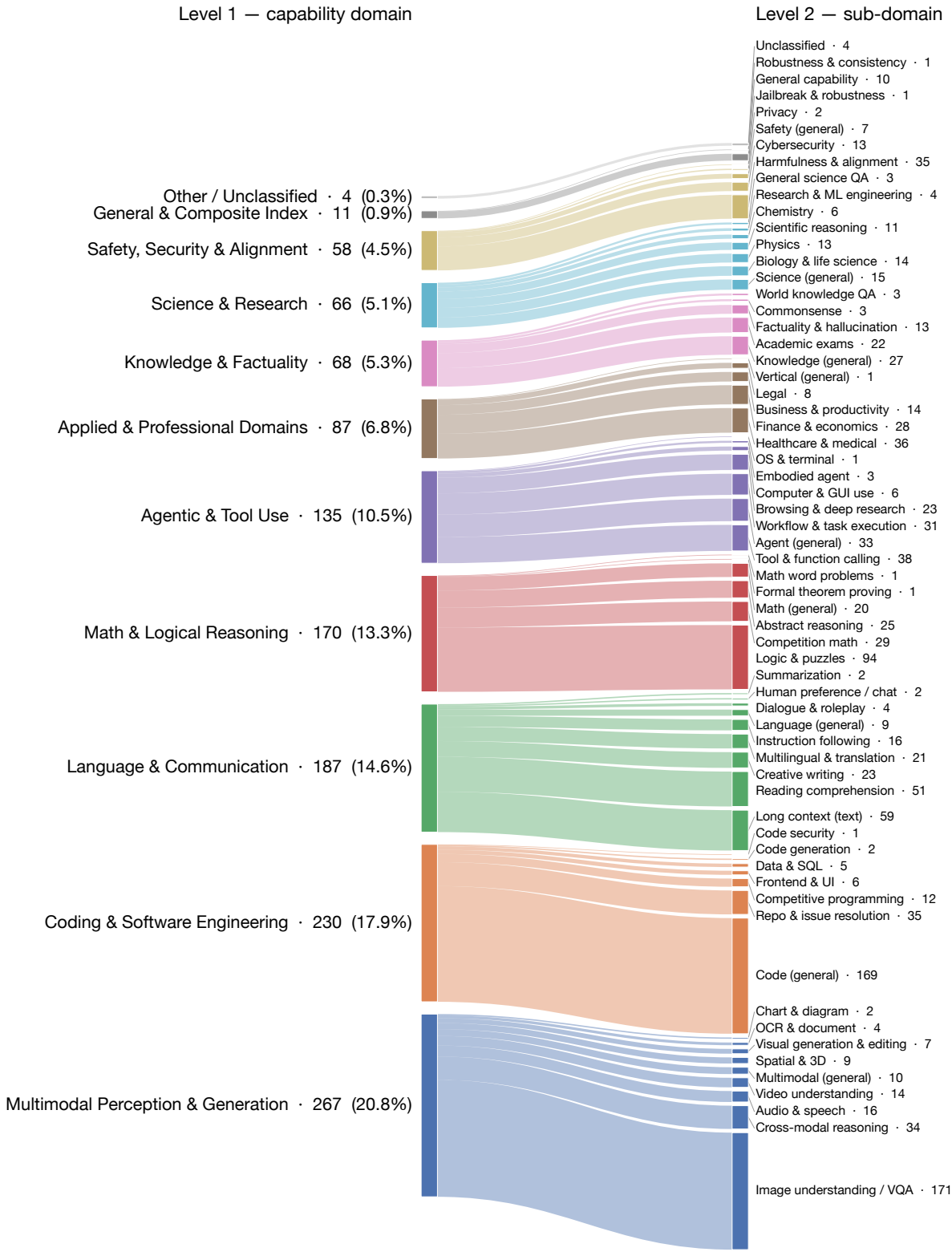



Figure 5: Capability taxonomy over the full catalog. Ribbon height is one unit per source record, so both columns carry all 1,283 records. Level 1 is one primary label per record.

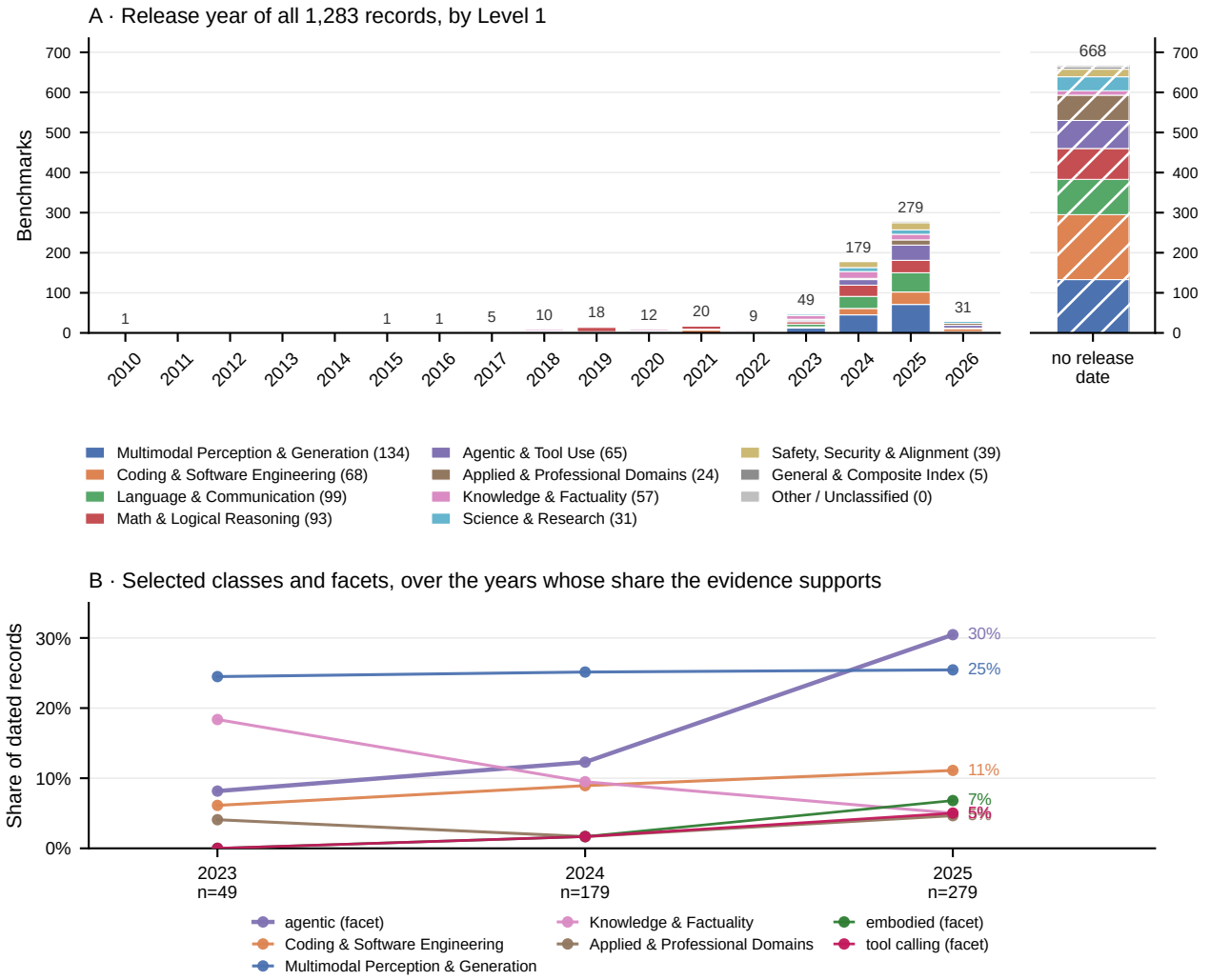



Figure 6: Release-year composition of the classified catalog. (A) The 615 dated source records by release year and Level 1 domain; the 668 records without a release date form the hatched column, drawn on the same scale. (B) Shares of selected domains and facets among each year's dated records, for years with at least 30 dated records and a source mix near the pooled mix. Panel A also shows the 108 dated records from the other years. Series marked *facet* cross the Level 1 classes and can overlap.

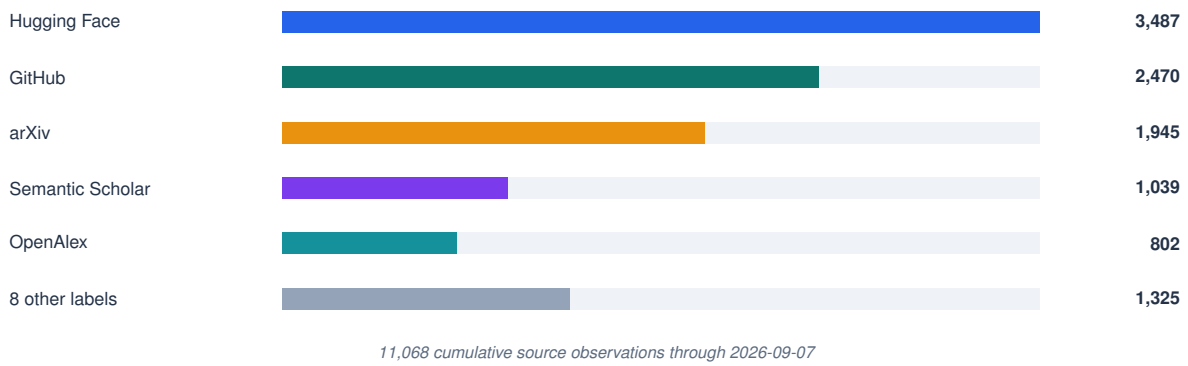



Figure 7: Discovery observations by source through 2026-09-07. Repeat sightings count as observations.




Figure 8: Discovery trends through September 7, 2026. Category cards show newly surfaced releases and track updates separately; the bars show daily evidence and attention volume. A release can carry several category tags. The visible coverage notice reports failed sources. The image links to the interactive Trends view.

## 5 Worked Example: Checking Prior Art

Before designing a new evaluation, a contributor surveyed August work on credit assignment in agentic training, with small Qwen-series models as a requirement for reproducible baselines. A coding agent installed the Benchmark Radar client and its public Skill, downloaded the corpus, and searched locally. It inspected the recorded paper, repository, and dataset links, tried additional web searches for work described in different terms, and read the source evidence before assembling the related-work table in Table 4.

The contributor used the table to assess whether the proposed evaluation duplicated existing work. Appendix E retains the session screenshots and an earlier manually assembled comparison.

| # | Work | Published | Qwen base model(s) | Credit-assignment focus | Agentic benchmarks used | Link |
|---|---|---|---|---|---|---|
| 1 | SRPO: Self-Reflective Policy Optimization for Long-Horizon Reasoning | 2026-08-25 | Qwen3-8B (also Qwen3-1.7B / 32B scaling) | Reflection-conditioned dense token-level signals convert sparse terminal reward into training signal | AIME'24, WebShop, ALFWorld, SWE-Bench-Lite | https://arxiv.org/abs/2608.23493 https://github.com/Galleons2029/SRPO |
| 2 | ContextPilot: Teaching Agents for Proactive Context Management via Fine-grained RL | 2026-08-28 | Qwen3-8B, Qwen3-14B | Coarse-grained → action-level credit assignment over branched trajectories during agentic RL | InfBench (InfiniteBench), NovelQA, LongMemEval, BrowseComp+ | https://arxiv.org/abs/2608.28476 https://github.com/Tencent/ContextPilot |
| 3 | SkillGate: Training In-Policy Skill Selection in Long-Horizon Agents | 2026-08-21 | Qwen3.5-9B (SFT checkpoint, RL init) | Selector credit starvation; partitions token support into two credit channels (outcome vs. action-local advantage) | 5 agentic benchmarks incl. SkillsBench, SWE (16-candidate skill slate) | https://arxiv.org/abs/2608.18852 |
| 4 | CIPO: Contextual Information Policy Optimization for Search Agents | 2026-08-06 | Qwen2.5-3B-Instruct, Qwen2.5-7B-Instruct | Dense turn-level credit (EALR) to evidence-using reasoning actions, combined with outcome reward | HotpotQA, 2WikiMultiHopQA, MuSiQue, Bamboogle + 3 OOD | https://arxiv.org/abs/2608.06128 |
| 5 | MoRSE: Task-Oriented Multi-Agent System with Mixture of Role-Subtask Experts | 2026-08-10 | Qwen3-4B-Instruct (one of three backbones; also Llama-3.1-8B, Gemma-4-31B) | Hierarchical GRPO with two-layer credit assignment (isolates expert vs. routing quality) | Code-generation benchmarks (multi-agent), held-out task domains | https://arxiv.org/abs/2608.09251 |

Table 4: Summary table of recent work on credit assignment in agentic training, assembled during the session. Each row links to its arXiv record and, where available, its repository.

## 6 Limitations and Future Work

The census describes this catalog at its recorded cutoff. Collection limits, failed requests, missing identifiers, and different snapshot dates affect its coverage. One underlying test can appear as several source records. Broader coverage requires additional source collection and review of the evidence already present.

Retrieval precision, task suitability, and time saved remain to be evaluated. The worked example combines local queries with web search and has no controlled baseline. Lexical matching can miss paraphrases and renamed tasks. Evaluating semantic retrieval [51] will require reviewed relevance judgments across queries and candidate records, including less familiar benchmark families.

The arXiv discovery route collects new submissions only, so it misses later versions of papers first posted before the collection window. The searches used for Section 2 missed two relevant earlier studies [1, 75], which a coauthor identified and read.

Measuring benchmark saturation or score stagnation requires comparable test versions and settings, with dates tied to score reporting or evaluation.

Classifying the 5,863 discovery-derived tracks by capability, and evaluating those labels, is future work.

## 7 Conclusion

Benchmark Radar brings daily benchmark discovery, catalog search, model-report mentions, and score histories into one living search engine. Readers can find candidate evaluations, locate task materials, inspect reporting choices, and follow the settings and citations behind a score. Shared benchmark IDs connect the web dashboard, downloadable catalog, and offline query clients.

The v0.11.0 release makes 1,283 source records searchable and preserves 464 unscored records with artifact links. Its full-catalog analyses distinguish benchmark records, scored models, cited documents, and numeric observations.

## Author Contributions

Koutian Wu led the work and manuscript preparation, built the initial collection pipeline, and implemented data aggregation across sources. Junjie Zhou prepared the score-archive audit and contributed report revisions. Wenjun Zhang contributed discussion and improvements to Benchmark Radar. Ergan Shang prepared figures and contributed copyediting. Jiayu Wang contributed the worked use case and its supporting evidence. Pengqian Han contributed data analysis and copyediting. Junkai Wang and Wanghan Xu contributed review and copyediting. Songyuanyi Lu contributed discussion and improvements to Benchmark Radar. Rudy Mizrahi Celekli, Lazizbek Ravshanov, and Yaoliang Bian contributed to software development for Benchmark Radar. Lin Shi contributed to conceptualization and methodology, and advised the work.

## Acknowledgements

We thank Ruohan Xu, Yifan Xu, Ran Hong, Hejia Geng, Haozhe Zhang, and Zhenxin Huang for discussion and for improving Benchmark Radar.

## References

[1] Mubashara Akhtar, Anka Reuel, Prajna Soni, Sanchit Ahuja, Pawan Sasanka Ammanamanchi, Ruchit Rawal, Vilém Zouhar, Srishti Yadav, Chenxi Whitehouse, Dayeon Ki, Jennifer Mickel, Leshem Choshen, Marek Šuppa, Jan Batzner, Jenny Chim, Jeba Sania, Yanan Long, Hossein A. Rahmani, Christina Knight, Yiyang Nan, Jyoutir Raj, Yu Fan, Shubham Singh, Subramanyam Sahoo, Eliya Habba, Usman Gohar, Siddhesh Pawar, Robert Scholz, Arjun Subramonian, Jingwei Ni, Mykel Kochenderfer, Sanmi

Koyejo, Mrinmaya Sachan, Stella Biderman, Zeerak Talat, Avijit Ghosh, and Irene Solaiman. When AI benchmarks plateau: A systematic study of benchmark saturation, 2026. URL https://arxiv.org/abs/2602.16763. Published at ICML 2026.

[2] Algolia. Hacker News Search API, 2026. URL https://hn.algolia.com/api. Accessed 2026-09-06.

[3] Anthropic. Introducing Claude Fable 5.1 and Claude Mythos 5.1, 2026. URL https://www.anthropic.com/claude-fable-and-mythos-5-1. Accessed 2026-09-07.

[4] Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31(3): 337–351, 2023. doi: 10.1017/pan.2023.2.

[5] Artificial Analysis. Artificial analysis intelligence benchmarking methodology, 2026. URL https://artificialanalysis.ai/methodology/intelligence-benchmarking. Accessed 2026-09-06.

[6] arXiv. arXiv API User Manual, 2026. URL https://info.arxiv.org/help/api/user-manual.html. Accessed 2026-09-06.

[7] Jonathan Bragg, Mike D'Arcy, Nishant Balepur, Dan Bareket, Bhavana Dalvi, Sergey Feldman, Dany Haddad, Jena D Hwang, Peter Jansen, Varsha Kishore, et al. AstaBench: Rigorous benchmarking of AI agents with a scientific research suite. *arXiv preprint arXiv:2510.21652*, 2025.

[8] Brave Search. Brave Search API Documentation, 2026. URL https://api-dashboard.search.brave.com/app/documentation/web-search/get-started. Accessed 2026-09-06.

[9] Jun Shern Chan, Neil Chowdhury, Oliver Jaffe, James Aung, Dane Sherburn, Evan Mays, Giulio Starace, Kevin Liu, Leon Maksin, Tejal Patwardhan, et al. MLE-Bench: Evaluating machine learning agents on machine learning engineering. In *International Conference on Learning Representations*, volume 2025, pages 50466–50494, 2025.

[10] Hui Chen, Miao Xiong, Yujie Lu, Wei Han, Ailin Deng, Yufei He, Jiaying Wu, Yibo Li, Yue Liu, and Bryan Hooi. MLR-Bench: Evaluating AI agents on open-ended machine learning research. *Advances in Neural Information Processing Systems*, 38, 2026.

[11] Citation File Format developers. Citation file format 1.2.0, 2021. URL https://citation-file-format.github.io/.

[12] Crossref. Crossref REST API, 2026. URL https://api.crossref.org. Accessed 2026-09-06.

[13] Haotian Cui, Chloe Wang, Hassaan Maan, Kuan Pang, Fengning Luo, Nan Duan, and Bo Wang. scGPT: Toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods*, 21: 1470–1480, 2024. doi: 10.1038/s41592-024-02201-0.

[14] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daume, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92, 2021. doi: 10.1145/3458723.

[15] Hejia Geng, Zesen Huang, Haoyang Li, Wenbin Li, Koutian Wu, Zihan Zhou, Yuanbo Pang, Weihao Liu, Zigong Xu, Zhiping Li, Zongzheng Zhang, Chuanfei Dong, Jiankai Sun, Tianzhe Zheng, Fengyu Xie, Yue Ma, Yueheng Shi, Tong Xie, Zonglin Di, Xianrong Liu, Qucheng Gao, Yimin Liu, Jiaming Pan, Sheng Huang, Xiao-Han Ma, Lanqing Yuan, Zhenlin Zhu, Ziang Liu, Ziyang Xu, Junkai Wang, Kangkai Liang, Jiayi Xian, Zehong Zhao, Liuwei Xu, Jingxu Xie, Peijin Zhang, Qiang Gao, Chengyi Xing, Zhe Zhao, Xi Wang, Yaopeng Xing, Xing Meng, Zhenfei Yin, Yingcheng Wu, and Ling Yang. ScienceIDE: Turning world's scientific codebase into agent learnable environments, 2026. URL https://arxiv.org/abs/2609.19134.

[16] Fabrizio Gilardi, Meysam Alizadeh, and Mael Kubli. ChatGPT outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, 120(30):e2305016120, 2023. doi: 10.1073/pnas.2305016120.

[17] Sankalp Gilda and Shlok Gilda. Position: Evaluation scores are perishable knowledge claims, 2026. URL https://arxiv.org/abs/2607.26191.

[18] GitHub Docs. REST API Endpoints for Releases, 2026. URL https://docs.github.com/en/rest/releases/releases. Accessed 2026-09-06.

[19] GitHub Docs. REST API Endpoints for Search, 2026. URL https://docs.github.com/en/rest/search/search. Accessed 2026-09-06.

[20] Harbor Contributors. Harbor: A framework for evaluating and improving agents, 2026. URL https://github.com/harbor-framework/harbor. Accessed 2026-09-10.

[21] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. *International Conference on Learning Representations*, 2021. URL https://arxiv.org/abs/2009.03300.

[22] Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. *arXiv preprint arXiv:2103.03874*, 2021. URL https://arxiv.org/abs/2103.03874.

[23] Tianyu Hua, Harper Hua, Violet Xiang, Benjamin Klieger, Sang Truong, Weixin Liang, Fan-Yun Sun, and Nick Haber. ResearchCodeBench: Benchmarking LLMs on implementing novel machine learning research code. *Advances in Neural Information Processing Systems*, 38, 2026.

[24] Hugging Face. Hugging Face Hub Documentation, 2026. URL https://huggingface.co/docs/hub/index. Accessed 2026-09-06.

[25] Pranab Islam, Anand Kannappan, Douwe Kiela, Rebecca Qian, Nino Scherrer, and Bertie Vidgen. FinanceBench: A new benchmark for financial question answering. *arXiv preprint arXiv:2311.11944*, 2023. URL https://arxiv.org/abs/2311.11944.

[26] Naman Jain, King Han, Alex Gu, Wen-Ding Li, Fanjia Yan, Tianjun Zhang, Sida Wang, Armando Solar-Lezama, Koushik Sen, and Ion Stoica. LiveCodeBench: Holistic and contamination free evaluation of large language models for code. *arXiv preprint arXiv:2403.07974*, 2024. URL https://arxiv.org/abs/2403.07974.

[27] Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can language models resolve real-world GitHub issues? *arXiv preprint arXiv:2310.06770*, 2023. URL https://arxiv.org/abs/2310.06770.

[28] Kaggle. Kaggle API Documentation, 2026. URL https://www.kaggle.com/docs/api. Accessed 2026-09-06.

[29] Wang-Cheng Kang and Julian McAuley. Self-attentive sequential recommendation. In *2018 IEEE International Conference on Data Mining*, pages 197–206, 2018. doi: 10.1109/ICDM.2018.00035.

[30] Daniel S. Katz et al. Recognizing the value of software: a software citation guide. *F1000Research*, 9:1257, 2021. doi: 10.12688/f1000research.26932.2. URL https://doi.org/10.12688/f1000research.26932.2.

[31] Rodney Michael Kinney, Chloe Anastasiades, Russell Authur, Iz Beltagy, Jonathan Bragg, Alexandra Buraczynski, Isabel Cachola, Stefan Candra, Yoganand Chandrasekhar, Arman Cohan, et al. The semantic scholar open data platform. *arXiv preprint arXiv:2301.10140*, 2023. URL https://arxiv.org/abs/2301.10140.

[32] Patrick Tser Jern Kon, Jiachen Liu, Xinyi Zhu, Qiuyi Ding, Jingjia Peng, Jiarong Xing, Yibo Huang, Yiming Qiu, Jayanth Srinivasa, Myungjin Lee, et al. EXP-Bench: Can AI conduct AI research experiments? *arXiv preprint arXiv:2505.24785*, 2025.

[33] Noam Koren, Roy Bar-Haim, and Abigail Goldsteen. Benchmarking the benchmarks: Evaluating benchmarks for conversational agents, 2026. URL https://arxiv.org/abs/2608.06329.

[34] Jon M. Laurent, Joseph D. Janizek, Michael Ruzo, Michaela M. Hinks, Michael J. Hammerling, Siddharth Narayanan, Manvitha Ponnapati, Andrew D. White, and Samuel G. Rodriques. LAB-Bench: Measuring capabilities of language models for biology research. *arXiv preprint arXiv:2407.10362*, 2024. URL https://arxiv.org/abs/2407.10362.

[35] Jure Leskovec, Daniel Huttenlocher, and Jon Kleinberg. Signed networks in social media. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, pages 1361–1370, 2010. doi: 10.1145/1753326.1753532.

[36] Xiaomin Li, Yuexing Hao, Jianheng Hou, Jintao Huang, Qianfeng Wen, Shirley Huang, Yifan Liu, Xiaoyi Liu, Yilan Fan, Yijun Wang, et al. MatrAIx: Simulating the world with 8.3 billion persona agents. *arXiv preprint arXiv:2608.04205*, 2026.

[37] Zongxia Li, Zhongzhi Li, Yucheng Shi, Ruhan Wang, Junyao Yang, Zhichao Liu, Xiyang Wu, Anhao Li, Yue Yu, Ninghao Liu, et al. Long-Horizon-Terminal-Bench: Testing the limits of agents on long-horizon terminal tasks with dense reward-based grading. *arXiv preprint arXiv:2607.08964*, 2026.

[38] Jhen-Ke Lin. Balance of benchmarks: Semantic density reweighting for benchmark multiplicity and task-conditioned evaluation, 2026. URL https://arxiv.org/abs/2608.30044.

[39] LLM Stats. AI and LLM benchmarks 2026: Rankings, scores and results, 2026. URL https://llm-stats.com/benchmarks/. Accessed 2026-09-06.

[40] Alisia Lupidi, Bhavul Gauri, Thomas Simon Foster, Bassel Al Omari, Despoina Magka, Alberto Pepe, Alexis Audran-Reiss, Muna Aghamelu, Nicolas Baldwin, Lucia Cipolina-Kun, et al. AIRS-Bench: A suite of tasks for frontier AI research science agents. *arXiv preprint arXiv:2602.06855*, 2026.

[41] Mike A. Merrill et al. Terminal-Bench: Benchmarking agents on hard, realistic tasks in command line interfaces, 2026. URL https://arxiv.org/abs/2601.11868.

[42] Adrian Mirza et al. Are large language models superhuman chemists? *arXiv preprint arXiv:2404.01475*, 2024. URL https://arxiv.org/abs/2404.01475.

[43] Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*, pages 220–229, 2019. doi: 10.1145/3287560.3287596.

[44] Deepak Nathani, Lovish Madaan, Nicholas Roberts, Nikolay Bashlykov, Ajay Menon, Vincent Moens, Amar Budhiraja, Despoina Magka, Vladislav Vorotilov, Gaurav Chaurasia, et al. MLGym: A new framework and benchmark for advancing AI research agents. *arXiv preprint arXiv:2502.14499*, 2025.

[45] Sky Ng, Brihi Joshi, Ishan Gupta, Shirley Huang, Zonglin Di, Yun Shen, Qianfeng Wen, Yifan Simon Liu, Ruoqi Gao, Zhiwei Zhang, et al. MicroVerse: An instrument for measuring self-authored identity drift in long-horizon multi-agent language-model simulations. *arXiv preprint arXiv:2608.15844*, 2026.

[46] OpenAI. GPT-6 Astra model, 2026. URL https://developers.openai.com/api/docs/models/gpt-6-astra. Accessed 2026-09-07.

[47] OpenCompass Contributors. Dataset statistics, 2026. URL https://opencompass.readthedocs.io/en/latest/dataset_statistics.html. Accessed 2026-09-06.

[48] OpenReview. OpenReview Documentation, 2026. URL https://docs.openreview.net/. Accessed 2026-09-06.

[49] Long Phan et al. Humanity's last exam. *arXiv preprint arXiv:2501.14249*, 2025. URL https://arxiv.org/abs/2501.14249.

[50] Jason Priem, Heather Piwowar, and Richard Orr. OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. *arXiv preprint arXiv:2205.01833*, 2022. URL https://arxiv.org/abs/2205.01833.

[51] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 3982–3992, 2019. doi: 10.18653/v1/D19-1410.

[52] David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. GPQA: A graduate-level google-proof Q&A benchmark. *arXiv preprint arXiv:2311.12022*, 2023. URL https://arxiv.org/abs/2311.12022.

[53] Stephen Robertson, Hugo Zaragoza, and Michael Taylor. Simple BM25 extension to multiple weighted fields. In *Proceedings of the Thirteenth ACM International Conference on Information and Knowledge Management*, 2004. doi: 10.1145/1031171.1031181.

[54] Nyamtulla Shaik, Fengjun Li, and Bo Luo. Benchmarking the benchmarks: Evaluating automated safety benchmarks for small language models, 2026. URL https://arxiv.org/abs/2608.17183.

[55] Ergan Shang and Flavio Sales Truzzi. ERASE: EaRly bAckpropagation SchEdule for Faster Training of Modern Recommendation Systems. *arXiv preprint arXiv:2608.18469*, 2026.

[56] Ergan Shang, Yuting Wei, and Kathryn Roeder. Predicting the unseen: a diffusion-based debiasing framework for transcriptional response prediction at single-cell resolution. *Proceedings of the National Academy of Sciences*, 122(52):e2525268122, 2025. doi: 10.1073/pnas.2525268122. URL https://doi.org/10.1073/pnas.2525268122.

[57] Ergan Shang, Weijing Tang, and Yinqiu He. LLM Evaluation on Unseen Questions: Contextual Multidimensional IRT Model. *arXiv preprint arXiv:2608.22295*, 2026.

[58] Ergan Shang, Yuan Zhang, and Weijing Tang. Inference for Balance in Dynamic Signed Networks. *arXiv preprint arXiv:2606.08786*, 2026.

[59] Lin Shi, Haowei Lin, Zixuan Zhu, Xiaoyue Zhou, Xiang Li, et al. Harbor adapters and Harbor-Index: Infrastructure and a curated meta-dataset for large-scale agentic evaluation, 2026. URL https://arxiv.org/abs/2609.04298.

[60] Philipp D. Siedler and Jordan Sassoon. Benchmarks are not monolithic: Sample-level auditing and orchestration for LLM evaluation, 2026. URL https://arxiv.org/abs/2607.28801.

[61] Arfon M. Smith, Daniel S. Katz, and Kyle E. Niemeyer. Software citation principles. *PeerJ Computer Science*, 2:e86, 2016. doi: 10.7717/peerj-cs.86. URL https://doi.org/10.7717/peerj-cs.86.

[62] Giulio Starace, Oliver Jaffe, Dane Sherburn, James Aung, Jun Shern Chan, Leon Maksin, Rachel Dias, Evan Mays, Benjamin Kinsella, Wyatt Thompson, et al. Paperbench: Evaluating ai's ability to replicate ai research. *arXiv preprint arXiv:2504.01848*, 2025.

[63] Fei Sun, Jun Liu, Jian Wu, Changhua Pei, Xiao Lin, Wenwu Ou, and Peng Jiang. BERT4Rec: Sequential recommendation with bidirectional encoder representations from transformer. In *Proceedings of the 28th ACM International Conference on Information and Knowledge Management*, pages 1441–1450, 2019. doi: 10.1145/3357384.3357895.

[64] Qiushi Sun, Zhoumianze Liu, Chang Ma, Zichen Ding, Fangzhi Xu, Zhangyue Yin, Haiteng Zhao, Zhenyu Wu, Kanzhi Cheng, Zhaoyang Liu, et al. ScienceBoard: Evaluating multimodal autonomous agents in realistic scientific workflows. *arXiv preprint arXiv:2505.19897*, 2025.

[65] Christina V. Theodoris et al. Transfer learning enables predictions in network biology. *Nature*, 618: 616–624, 2023. doi: 10.1038/s41586-023-06139-9.

[66] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL https://arxiv.org/abs/1706.03762.

[67] Xiaoxuan Wang, Ziniu Hu, Pan Lu, Yanqiao Zhu, Jieyu Zhang, Satyen Subramaniam, Arjun R. Loomba, Shichang Zhang, Yizhou Sun, and Wei Wang. SciBench: Evaluating college-level scientific problem-solving abilities of large language models. *arXiv preprint arXiv:2307.10635*, 2023. URL https://arxiv.org/abs/2307.10635.

[68] Kyle Waters, Lucas Nuzzi, Tadhg Looram, Alessandro Tomasiello, Ariel Ghislain Kemogne Kamdoum, Bikun Li, Damien Sileo, Egor Kretov, Francesco Fournier-Facio, Georgios Soloupis, et al. Composite-stem. *arXiv preprint arXiv:2604.09836*, 2026.

[69] Koutian Wu, Wen Yi, Xianghui Xue, Iain Murray Reid, and Maolin Lu. Diurnal and seasonal variations of meteor speed and arrival angle observed by mengcheng meteor radar. *Journal of Geophysical Research: Space Physics*, 129(10):e2024JA032767, 2024. doi: 10.1029/2024JA032767. URL https://doi.org/10.1029/2024JA032767.

[70] Koutian Wu, Yuxuan Cao, Gengchen Mai, and Zeping Liu. ESM-Bench: A benchmark for evaluating whether AI agents understand earth system model physics and code. In *HydroML 2026 Conference Poster*. HydroML, 2026. URL https://zenodo.org/records/19802836.

[71] Yunze Wu, Dayuan Fu, Weiye Si, Zhen Huang, Mohan Jiang, Keyu Li, Shijie Xia, Jie Sun, Tianze Xu, Xiangkun Hu, et al. Innovatorbench: Evaluating agents' ability to conduct innovative llm research. *arXiv preprint arXiv:2510.27598*, 2025.

[72] L. Xiaopai. Builderpulse: Ai-powered daily intelligence for indie hackers and builders. GitHub, 2026. URL https://github.com/BuilderPulse/BuilderPulse.

[73] Shiyun Xiong, Dongming Wu, Peiwen Sun, Yuang Ai, Bokang Yang, Wencheng Han, Xiao-Hui Li, and Xiangyu Yue. Benchmark Everything Everywhere All at Once, 2026. URL https://arxiv.org/abs/2606.06462.

[74] Wanghan Xu, Shuo Li, Tianlin Ye, Qinglong Cao, Yixin Chen, Hengjian Gao, Yiheng Wang, Qi Li, Kun Li, Sheng Xu, et al. ResearchClawBench: A benchmark for end-to-end autonomous scientific research. *arXiv preprint arXiv:2606.07591*, 2026.

[75] Eddie Yang and Dashun Wang. Benchmark illusion: Disagreement among LLMs and its scientific consequences, 2026. URL https://arxiv.org/abs/2602.11898.

[76] Zenodo. Zenodo REST API Documentation, 2026. URL https://developers.zenodo.org/. Accessed 2026-09-06.

[77] Tianyu Zhang, Ergan Shang, and Kathryn Roeder. Genetic Convergence Analysis of CRISPR Perturbations Deciphers Gene Functional Similarity. *bioRxiv*, 2025. doi: 10.1101/2025.11.13.688060. URL https://doi.org/10.1101/2025.11.13.688060.

[78] Junwei Zhou, Zhen Sun, Binyu Li, Jiangyu Zhou, Yuexi Pan, Hengyu Wang, Honghe Ren, Xiaohan Jia, Xueyang Zhou, Xiaoyu Cao, et al. ASI-Bench: At the Dawn of Artificial Superintelligence. *arXiv preprint arXiv:2608.17271*, 2026.

# Appendix

## A Full-Catalog Census

This census retains all 1,283 source records, including those without scores.

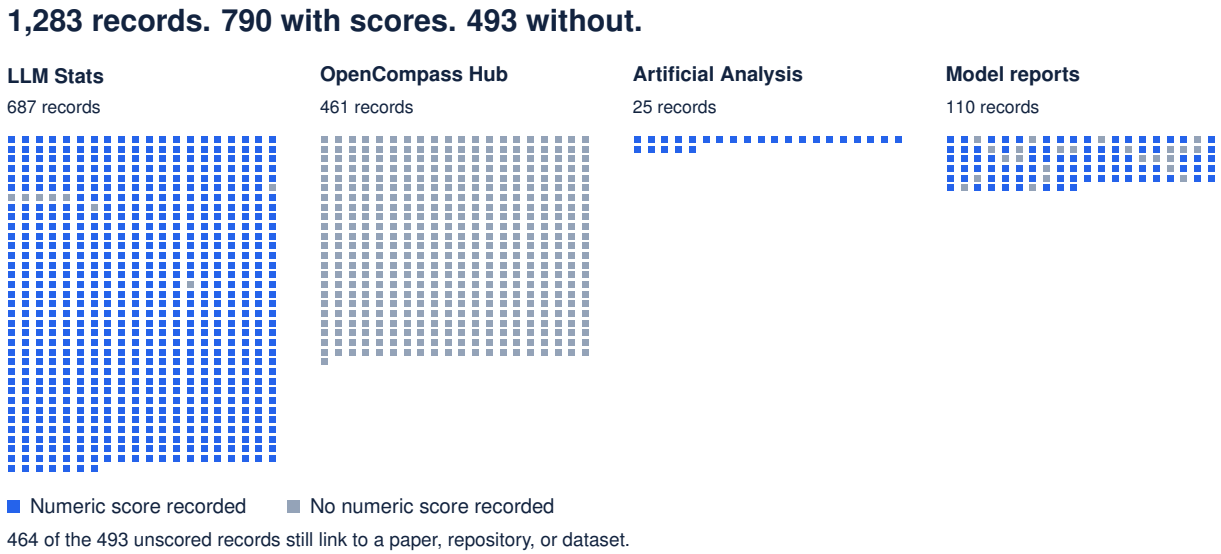



Figure 9: 464 unscored entries provide artifact links. Each mark represents one source record and links to its detail page. All 1,283 records appear.

## B Catalog Evidence and Measurement Coverage

The following analyses use the same catalog as Section 4. They document the evidence behind its records and the requirements for comparing measurements.

### B.1 Cited Documents and Scored Models

The common document registry contains 1,208 distinct cited documents, attached to 1,278 of the 1,283 benchmark records. 5 records have no cited documents. The shared model registry contains 868 model identities. Model identifiers are available for every numeric score in all 790 scored records, allowing distinct models to be counted within each record. For unscored records, this count is unknown.

Table 5 compares the three counts for GPQA Diamond. The Artificial Analysis record for GPQA Diamond contains scores for 586 models and cites 1 registry page. The Model reports record cites 27 documents and contains 21 numeric observations for 19 models.

| Benchmark record | Source | Models | Documents | Scores |
|---|---|---|---|---|
| GPQA Diamond | Artificial Analysis | 586 | 1 | 586 |
| GPQA Diamond | Model reports | 19 | 27 | 21 |
| Humanity's Last Exam | Artificial Analysis | 577 | 1 | 577 |
| SciCode | Artificial Analysis | 577 | 1 | 577 |
| CritPt | Artificial Analysis | 492 | 1 | 492 |

Table 5: One cited page can document hundreds of scored models. Models count distinct source model IDs within a record; documents count distinct citation IDs; scores count numeric observations.

Table 5 also records 577 scored models for Humanity's Last Exam, 577 for SciCode, and 492 for CritPt, each under one Artificial Analysis citation.

### B.2 Score Scales and Comparison Eligibility

Only 82 of the 790 scored records declare a percentage unit with a known direction and values within 0–100. The remaining 708 scored records use other or unverified scales. Table 6 accounts for the entire catalog.

| **Measurement state** | **Records** | **Interpretation** |
|---|---|---|
| Numeric scores, declared percentage scale | 82 | Supports scale-specific summaries. |
| Numeric scores, other or unverified scale | 708 | Scores kept on their native scale. |
| No numeric score | 493 | Score headroom unknown. |
| Full population | 1,283 | Every source record. |

Table 6: Only 82 of 790 scored records meet the percentage-scale rule in Section 3.5.

The catalog records benchmark release dates for 615 records, leaving 668 without a release date. Across score observations, 12,594 entries carry model-announcement dates and 322 carry document-publication dates.

### B.3 Documentation Across All Catalog Sources

The documentation analysis covers all 1,283 benchmark records. It finds 1,208 distinct cited documents supporting 1,278 records; 5 records have no cited document. Table 7 covers every source. Each distinct document counts once, whether it is a registry page or a model report.

| Source | Records | With docs | Documents | Docs without named org. |
|---|---|---|---|---|
| LLM Stats | 687 | 687 | 687 | 687 |
| OpenCompass Hub | 461 | 461 | 461 | 461 |
| Artificial Analysis | 25 | 25 | 23 | 23 |
| Model reports | 110 | 105 | 37 | 0 |
| Total | 1,283 | 1,278 | 1,208 | 1,171 |

Table 7: Documentation across the complete v0.11.0 catalog. "With docs" counts benchmark records; the last two columns count distinct documents.

The release data names no organization for 1,171 cited documents. The complete export retains each document's identity and source URL on its benchmark record.

### B.4 Reported Scores Across the Full Catalog

The score analysis covers 12,916 numeric observations. Table 8 summarizes scored-model coverage for each source. 22 benchmark records contain numeric scores for at least 100 distinct source model IDs.

| Source | Records | Scored | Unscored | Score rows | Median models | Max. models |
|---|---|---|---|---|---|---|
| LLM Stats | 687 | 679 | 8 | 5,544 | 3 | 239 |
| OpenCompass Hub | 461 | 0 | 461 | 0 | Unknown | Unknown |
| Artificial Analysis | 25 | 25 | 0 | 7,050 | 245 | 586 |
| Model reports | 110 | 86 | 24 | 322 | 2 | 19 |
| Total | 1,283 | 790 | 493 | 12,916 | | |

Table 8: Reported scores across every source in the v0.11.0 release. Models count distinct source model IDs within each benchmark record. Medians and maxima use records with known model counts. The record-level score export contains every catalog record, including raw maxima, all tied reporting models and citations, units, settings, and explicit missing values.

The accompanying `evidence/catalog-findings.csv` contains one row per benchmark record, with a JSON counterpart retaining detailed evidence. Each scored record has its highest numeric value on the source's native scale and every tied score observation. Units and score direction remain attached to that record; for a lower-is-better metric, the best result is the minimum.

Percentage headroom is 100 minus a record's highest value, computed when its unit is explicitly percent, its direction is higher-is-better, and its values stay within 0–100. The export records eligibility for every benchmark. Catalog-wide headroom and recency estimates report their eligible measurement coverage alongside the statistic. Comparing results also requires matching test versions, reasoning budgets, tools, attempts, and evaluators [14, 43].

### B.5 Date Evidence Across the Full Catalog

Table 9 separates benchmark release dates from dates attached to numeric observations. Both date bases are proxies for the evaluation date.

| Source | Records | Release known | Doc.-dated scores | Model-dated scores | Undated scores |
|---|---|---|---|---|---|
| LLM Stats | 687 | 42 | 0 | 5,544 | 0 |
| OpenCompass Hub | 461 | 457 | 0 | 0 | 0 |
| Artificial Analysis | 25 | 17 | 0 | 7,050 | 0 |
| Model reports | 110 | 99 | 322 | 0 | 0 |
| Total | 1,283 | 615 | 322 | 12,594 | 0 |

Table 9: Date evidence for the complete v0.11.0 catalog. The first two numeric columns count benchmark records; the final three count numeric score observations.

The exported rows preserve each record's release date and the date basis of its scores.

## C Collection Settings and Display Filters

The 2026-09-07 snapshot records 1,003 fetched rows and 954 candidates after duplicate removal. Of these, 366 entered the published snapshot and 138 met the recommendation threshold. These counts describe the discovery run. Published evidence retains source URLs, retrieval times, parser versions, identifiers, and payload checksums. Raw responses and credentials stay private.

Benchmark Frontier, on the Leaderboard tab, requires a numeric score and excludes known pre-2024 benchmarks. It retains scored records with unknown dates or counts through labelled marks, and keeps unverified scales outside Pareto calculations. Its initial score cutoff is 70. The linked score ranking keeps its own date and numeric-score requirements.

A separate priority score for daily recommendations combines relevance (35%), evidence (20%), recency (20%), and adoption (25%) on a 0–100 scale. These weights set the reading priority of discovery observations.

## D Source Inventory and Health

Benchmark Radar ingests from 37 sources: 13 direct connectors and 24 first-party feeds. The direct connectors read arXiv [6], Hugging Face Hub [24], GitHub Search and GitHub Releases [18, 19], Kaggle [28], Zenodo [76], Crossref [12], OpenAlex [50], OpenReview [48], Semantic Scholar [31], Brave Search [8], and Hacker News [2]. Table 10 lists source connectors that completed successfully at the cutoff. Row counts are measured before duplicate removal and ranking. A connector is healthy when its request succeeds and its response parses. Core sources cover papers, shared model or dataset artifacts, and code.

| Connector | Role | Cutoff status | Core |
|---|---|---|---|
| arXiv | Primary papers via selected AI, language, vision, and software categories | Healthy; 46 rows | Yes |
| Hugging Face Hub | Datasets and Spaces connected to benchmark artifacts | Healthy; 101 rows | Yes |
| GitHub Search | Code repositories and benchmark artifacts | Healthy; 300 rows; at cap | Yes |
| GitHub Organizations | Reviewed organization repositories | Healthy; 6 rows | No |
| Hugging Face Papers | Community-surfaced papers | Healthy; 26 rows | No |
| Kaggle Datasets | Public benchmark datasets | Healthy; 30 rows | No |
| Zenodo | DOI-bearing research artifacts | Healthy; 65 rows | No |
| Crossref | DOI metadata | Healthy; 180 rows | No |
| OpenReview | Conference submissions | Healthy; no eligible rows | No |
| GitHub Releases | First-party releases | Healthy; 1 row | No |
| OpenAlex | Scholarly discovery metadata | Healthy; 248 rows | No |

Table 10: Healthy direct discovery connectors at the current cutoff.

Table 11 lists research and engineering feeds operated by the institutions being monitored. They can surface benchmark papers, datasets, model announcements, and engineering posts. The feed collector returned no records at the cutoff. Semantic Scholar returned a malformed payload, and Brave Search lacked an API key. The snapshot records these failures. Searches restricted to an organization's website cover institutions that lack a verified feed.

| Feeds 1-12 | Feeds 13-24 |
|---|---|
| Meituan Engineering | Ai2 |
| OpenAI News | Together AI |
| Google AI | Sakana AI |
| Google DeepMind | Qwen |
| Google Research | Ollama |
| Apple Machine Learning Research | Stability AI |
| AWS Machine Learning | Nomic AI |
| Hugging Face Blog | Replicate |
| Microsoft Research | NVIDIA Developer |
| NVIDIA AI Blog | IBM Research |
| Mistral AI | Databricks |
| Meta Research | LangChain |

Table 11: The 24 first-party research and engineering feeds monitored by Benchmark Radar.

# E Worked Use Case: Prior-Art Check for a New Evaluation

The contributor wanted to determine whether a proposed evaluation duplicated existing work. They surveyed work published in August 2026 on credit assignment in agentic training, using small Qwen-series models as a requirement for reproducible baseline experiments. The search produced a focused comparison table that the contributor used to assess the proposed design.

The contributor gave the task to a coding agent together with the public setup instructions for Benchmark Radar. The agent installed the command-line client and the Benchmark Radar Skill, a set of instructions for using the client. It then downloaded the local corpus and queried candidate records offline (Figure 10).

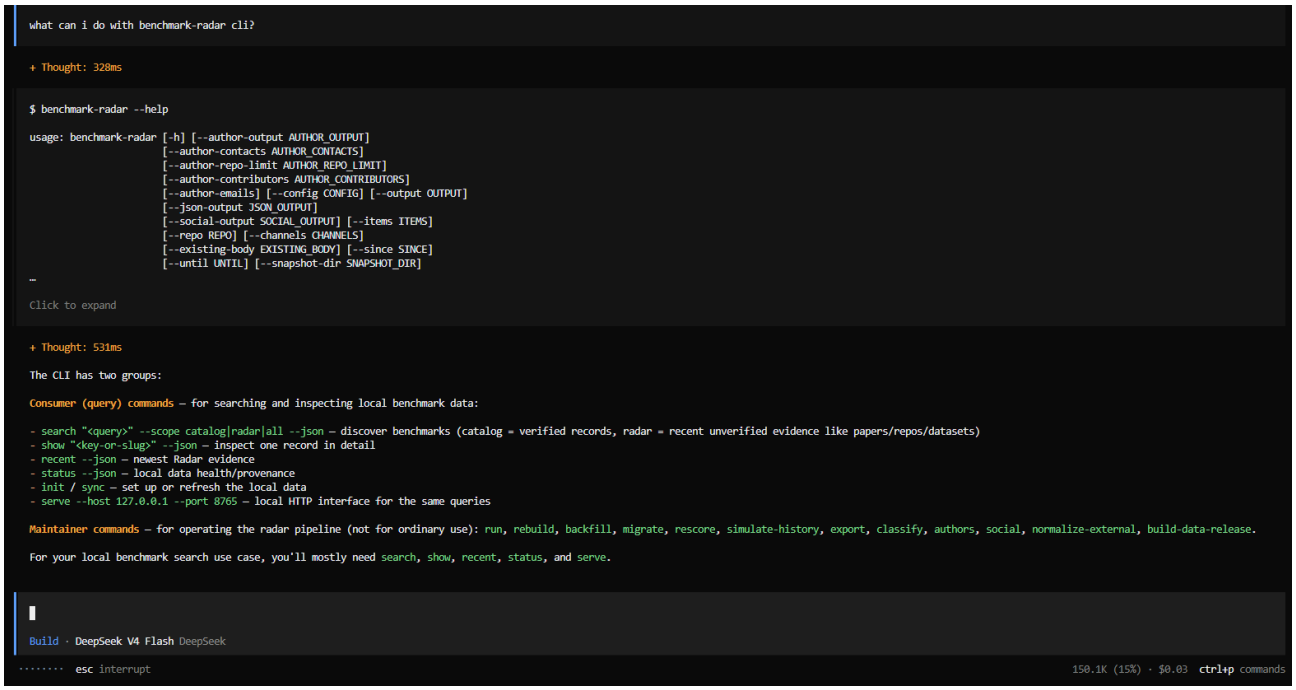



Figure 10: A coding agent installs the Benchmark Radar client and its usage instructions, then runs local queries.

The agent inspected paper, repository, and dataset links to decide which sources to open next. Figures 11 and 12 show two retrieved discovery records: one has a repository link, and the other has both paper and repository links.

The agent supplemented local queries with web search to find related work described in different terms. It cross-checked the candidate sets and inspected the source evidence before selecting records for the comparison. Figure 13 shows the word matches and score components available for inspecting one retrieved candidate.

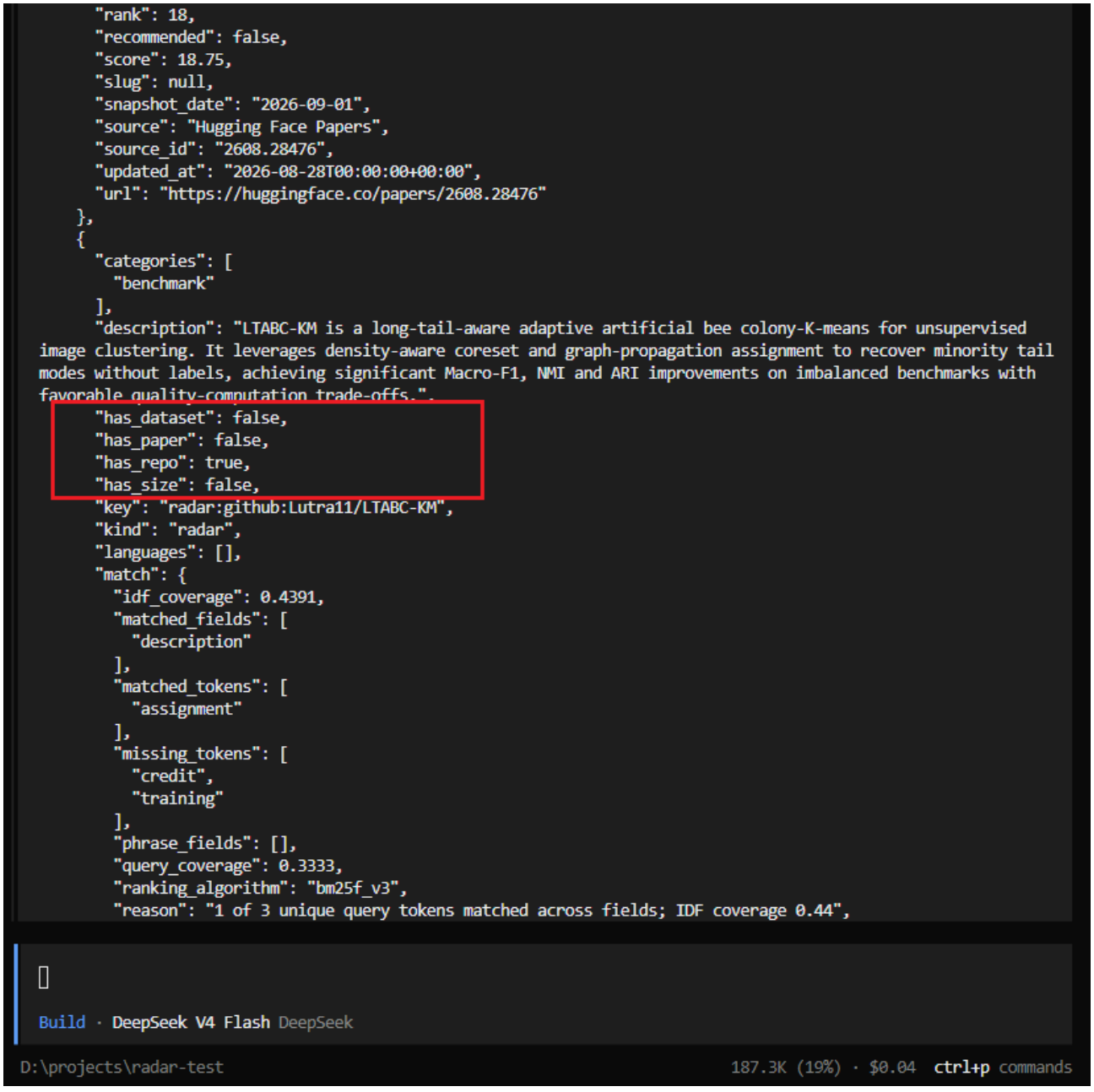



Figure 11: A retrieved discovery record with a repository link for inspecting the implementation.

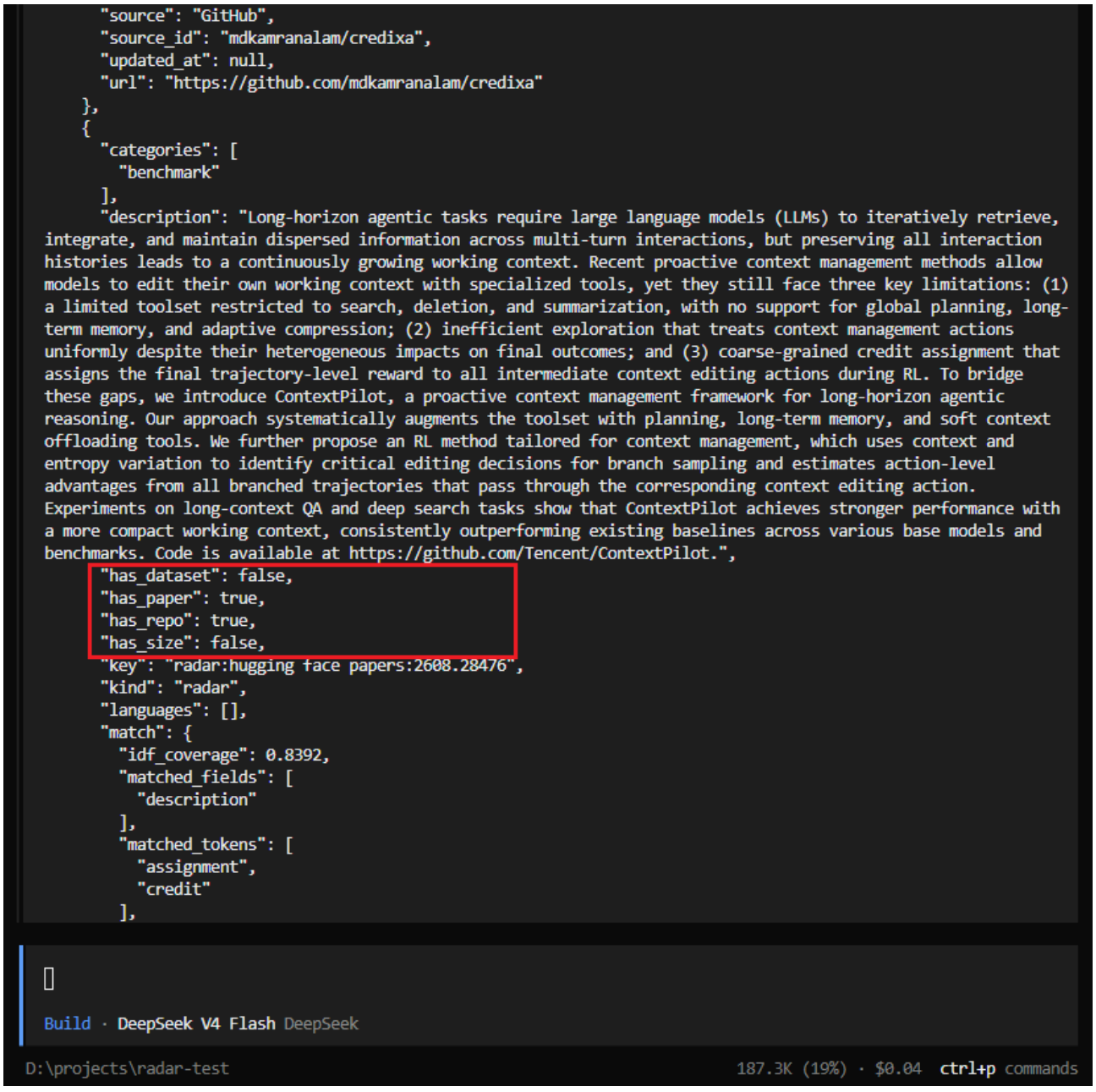



Figure 12: A second discovery record provides both paper and repository links.

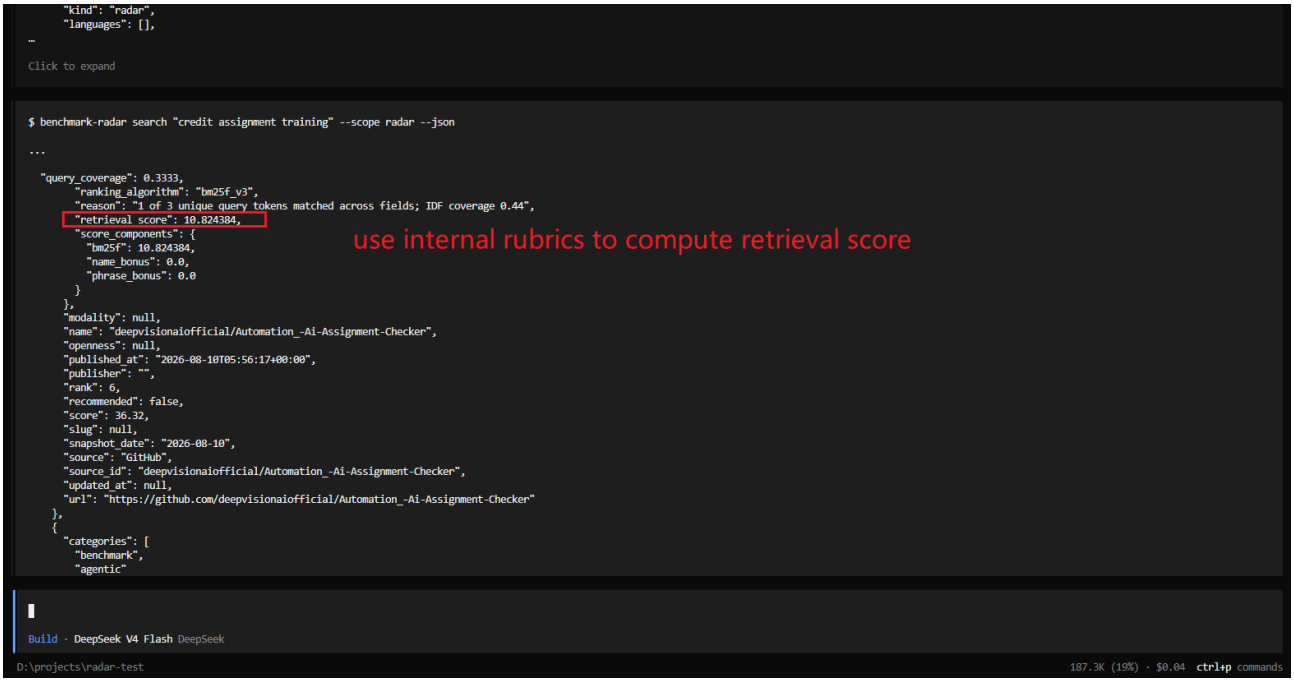



Figure 13: A candidate record with query-word matches and retrieval-score components for the agent to inspect when judging relevance.

The resulting related-work table appears in Table 4 in the main text.

Table 12 shows a prior-art comparison table from an earlier benchmark-design effort by one contributor. Each row is one related benchmark and each column one design dimension, with each cell read from a source paper. The contributor reported that assembling it took more effort than any part of that project except producing the benchmark data. Benchmark Radar retrieves candidates for this kind of comparison and exposes the matching words and fields for inspection.

| Bench Name | End-to-End Tasks | Fine-Grained Eval | Researcher-Quality Eval | Data Generation | Multi-Harness Eval | #Tasks |
|---|---|---|---|---|---|---|
| MLE-Bench [9] | ✗ | ✓ | ✗ | Transfer&Compose | ✓ | 75 |
| MLGym-Bench [44] | ✗ | ✓ | ✗ | Automatic | ✗ | 13 |
| EXP-Bench [32] | ✓ | ✗ | ✗ | Automatic | ✓ | 461 |
| ResearchCodeBench [23] | ✗ | ✓ | ✗ | Transfer&Compose | ✗ | 212 |
| MLR-Bench [10] | ✓ | ✗ | ✗ | Automatic | ✓ | 201 |
| PaperBench [62] | ✗ | ✓ | ✗ | Transfer&Compose | ✗ | 8316 |
| AstaBench [7] | ✓ | ✓ | ✗ | Transfer&Compose | ✓ | 2400+ |
| InnovatorBench [71] | ✓ | ✗ | ✗ | Transfer&Compose | ✗ | 20 |
| AIRS-Bench [40] | ✗ | ✓ | ✗ | Automatic | ✓ | 20 |
| COMPOSITE-Stem [68] | ✓ | ✓ | ✗ | Manual | ✗ | 70 |
| ScienceBoard [64] | ✗ | ✓ | ✗ | Manual | ✗ | 169 |

Table 12: A prior-art comparison table assembled manually for an earlier benchmark-design effort. Each row is one related benchmark and each column one design dimension, all read from the source papers by hand.

# F Reproducibility, Access, and Citation

All reported numbers come from Benchmark Radar release v0.11.0, with discovery data through 2026-09-07. Two audit scripts recompute the census and the document, score, and date analyses from that

release's catalog, registries, and dated snapshots. Capability classification and its time trends are generated separately and are not covered by these scripts.

The paper repository's reproduction guide identifies the scripts, input hashes, and validation procedure. Table 13 lists the access and citation resources. The project also provides a Citation File Format record [11, 30, 61].

| **Artifact** | **Canonical or permanent location** |
|---|---|
| Source code | https://github.com/ktwu01/benchmark-radar |
| Dashboard | https://benchmark-radar.org/ |
| Dataset (Hugging Face) | https://huggingface.co/datasets/ktwu01/benchmark-radar |
| RSS | https://benchmark-radar.org/feed.xml |
| Citation metadata | https://github.com/ktwu01/benchmark-radar/blob/main/CITATION.cff |

Table 13: Access and citation artifacts for Benchmark Radar.

## Licensing

The software is released under the MIT License. This paper and the project's original editorial content are released under CC BY-NC-SA 4.0. Third-party source material remains under its original terms.